

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(i)	States that $S = a + (a + d) + \dots + (a + (n - 1)d)$	B1	1.1a
	$S = a + (a + d) + (a + (n - 1)d)$ $S = (a + (n - 1)d) + (a + (n - 2)d) + \dots + a$	M1	3.1a
	Reaches $2S = n \times (2a + (n - 1)d)$ And so proves that $S = \frac{n}{2}[2a + (n - 1)d]$ *	A1*	2.1
		(3)	
(ii)	(a) $S = 10 + 9.20 + 8.40 + \dots$		
	$64 = \frac{n}{2}(20 - 0.8(n - 1))$ o.e	M1	3.1b
	$128 = 20n - 0.8n^2 + 0.8n$ $0.8n^2 - 20.8n + 128 = 0$ $n^2 - 26n + 160 = 0$ *	A1*	2.1
		(2)	
	(b) $n = 10, 16$	B1	1.1b
		(1)	
	(c) 10 weeks with a minimal correct reason. E.g. • He has saved up the amount by 10 weeks so he would not save for another 6 weeks • You would choose the smaller number • He starts saving negative amounts (in week 14) so 16 does not make sense	B1	2.3
		(1)	
(7 marks)			
Notes:			

(i)

B1: Correctly writes down an expression for the key terms S or S_n including $S =$ or $S_n =$

Allow a minimum of 3 correct terms including the first and last terms, and no incorrect terms.

Score for S or $S_n = a + (a + d) + \dots + (a + (n - 1)d)$ with + signs, not commas

If the series contains extra terms that should not be there E.g

$S = a + (a + d) + \dots + (a + nd) + (a + (n - 1)d)$ score B0

M1: For the key step in reversing the terms and adding the two series.

Look for a minimum of two terms, including a and $a + (n - 1)d$, the series reversed with evidence of adding, for example $2S =$ Condone the extra incorrect terms (see above) appearing.

Can be scored when terms are separated by commas

A1*: Shows correct work (no errors) with all steps shown leading to given answer.

There should be no incorrect terms. A minimum of 3 terms should be shown in each sum

The solution below is a variation of this.

$$S = a + (a+d) + \dots + l$$

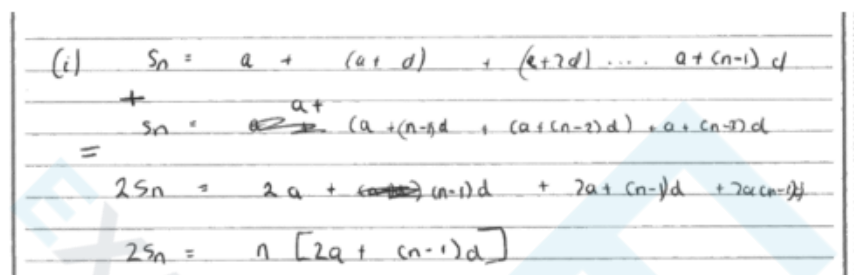
$$S = l + (l-d) + \dots + a$$

$$2S = n(a+l)$$

$$S = \frac{n}{2}(a+l) = \frac{n}{2}(a + a + (n-1)d) = \frac{n}{2}(2a + (n-1)d)$$

B1 and A1 are not scored until the last line, M scored on line 3

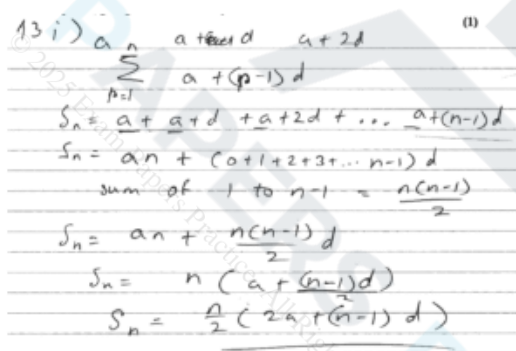
The following scores B1 M0 A0 as the terms in the second sum are not reversed



$$\begin{aligned}
 (i) \quad S_n &= a + (a+d) + (a+2d) \dots a + (n-1)d \\
 &+ \\
 S_n &= a + (a+(n-1)d) + (a+(n-2)d) + \dots + a + (n-1)d \\
 = \\
 2S_n &= 2a + (n-1)d + 2a + (n-1)d + \dots + 2a + (n-1)d \\
 2S_n &= n[2a + (n-1)d]
 \end{aligned}$$

SC in (a) Scores B1 M0 A0.

They use $0+1+2+\dots+(n-1) = \frac{n(n-1)}{2}$ which relies on the quoted proof.



$$\begin{aligned}
 (i) \quad S_n &= a + (a+d) + (a+2d) \dots a + (n-1)d \\
 &= \sum_{p=1}^n a + (p-1)d \\
 S_n &= a + a + d + a + 2d + \dots + a + (n-1)d \\
 S_n &= an + (0+1+2+\dots+n-1)d \\
 \text{sum of } 1 \text{ to } n-1 &= \frac{n(n-1)}{2} \\
 S_n &= an + \frac{n(n-1)}{2}d \\
 S_n &= n \left(a + \frac{(n-1)d}{2} \right) \\
 S_n &= \frac{n}{2} (2a + (n-1)d)
 \end{aligned}$$

(ii) (a)

M1: Uses the information given to set up a correct equation in n .

The values of S , a and d need to be correct and used within a correct formula

Possible ways to score this include unsimplified versions $64 = \frac{n}{2}(2 \times 10 + (n-1) \times -0.8)$,

$$64 = \frac{n}{2}(10+10+(n-1) \times -0.8) \text{ or versions using pence rather than £'s } 6400 = \frac{n}{2}(2000+(n-1) \times -80)$$

Allow recovery for both marks following $64 = \frac{n}{2}(2 \times 10 + (n-1) \times -0.8)$ with an invisible $\times$

A1*: Proceeds without error to the given answer. (Do not penalise a missing final trailing bracket)

Look for at least a line with the brackets correctly removed as well as a line with the terms in n correctly combined

$$\text{E.g. } 64 = \frac{n}{2}(20 + (n-1) \times -0.8) \Rightarrow 64 = 10n - 0.4n^2 + 0.4n \Rightarrow 0.4n^2 - 10.4n + 64 = 0 \Rightarrow n^2 - 26n + 160 = 0$$

(ii)(b)

B1: $n = 10, 16$

(ii)(c)

B1: Chooses 10 (weeks) and gives a minimal acceptable reason. The reason must focus on why the answer is 10 (weeks) rather than 16(weeks) or alternatively why it would not be 16 weeks.

Question	Scheme	Marks	AOs
(i)	$x^2 - 6x + 10 = (x-3)^2 + 1$	M1	2.1
	Deduces "always true" as $(x-3)^2 \geq 0 \Rightarrow (x-3)^2 + 1 \geq 1$ and so is always positive	A1	2.2a
		(2)	
(ii)	For an explanation that it need not (always) be true This could be if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$	M1	2.3
	States 'sometimes' and explains if $a > 0$ then $ax > b \Rightarrow x > \frac{b}{a}$ if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$	A1	2.4
		(2)	
(iii)	Difference $= (n+1)^2 - n^2 = 2n+1$	M1	3.1a
	Deduces "Always true" as $2n+1 = (\text{even} + 1) = \text{odd}$	A1	2.2a
		(2)	
(6 marks)			

Notes:

(i)

M1: Attempts to complete the square or any other valid reason. Allow for a graph of $y = x^2 - 6x + 10$ or an attempt to find the minimum by differentiation

A1: States always true with a valid reason for their method

(ii)

M1: For an explanation that it need not be true (sometimes). This could be if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$ or simply $-3x > 6 \Rightarrow x < -2$

A1: Correct statement (sometimes true) and explanation

(iii)

M1: Sets up the proof algebraically.

For example by attempting $(n+1)^2 - n^2 = 2n+1$ or $m^2 - n^2 = (m-n)(m+n)$ with $m = n+1$

A1: States always true with reason and proof

Accept a proof written in words. For example

If integers are consecutive, one is odd and one is even

When squared $\text{odd} \times \text{odd} = \text{odd}$ and $\text{even} \times \text{even} = \text{even}$

The difference between odd and even is always odd, hence always true

Score M1 for two of these lines and A1 for a good proof with all three lines or equivalent.

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$g(0) = 5$	M1	This mark is given for a method to find $g(0)$
	$gg(0) = g(5) = 13$	A1	This mark is given for a correct value for $gg(0)$
(b)	$(x-2)^2 + 1 > 28$ $(x-2)^2 > 27$ $x-2 > 3\sqrt{3}$	M1	This mark is given for a method to solve $g(x) > 28$ when $x \leq 2$
	$x < 2 - 3\sqrt{3}$	A1	
	$4x - 7 > 28$ $4x > 35$ $x > \frac{35}{4}$	M1	This mark is given for a solving $g(x) > 28$ when $x > 2$
	$x < 2 - 3\sqrt{3}$ and $x > \frac{35}{4}$	A1	This mark is given for a correct range of values of x for which $g(x) > 28$ stated
(c)	h^{-1} exists since h is a one-to-one function; g^{-1} does not exist since g is a many-to-one function	B1	This mark is given for a valid explanation
(d)	$h^{-1}(x) = 2 - \sqrt{x-1}$	B1	This mark is given for finding an expression for $h^{-1}(x)$
	$2 \pm \sqrt{x-1} = -\frac{1}{2}$	M1	This mark is given for a method to rearrange to find a value for x
	$x = 7.25$	A1	This mark is given for a correct value of x
(Total 10 marks)			

Q4.

Question	Scheme	Marks	AOs
(a) Way 1	$H = Ax(40 - x)$ {or $H = Ax(x - 40)$ }	M1	3.3
	$x = 20, H = 12 \Rightarrow 12 = A(20)(40 - 20) \Rightarrow A = \frac{3}{100}$	dM1	3.1b
	$H = \frac{3}{100}x(40 - x)$ or $H = -\frac{3}{100}x(x - 40)$	A1	1.1b
	(3)		
(a) Way 2	$H = 12 - \lambda(x - 20)^2$ {or $H = 12 + \lambda(x - 20)^2$ }	M1	3.3
	$x = 40, H = 0 \Rightarrow 0 = 12 - \lambda(40 - 20)^2 \Rightarrow \lambda = \frac{3}{100}$	dM1	3.1b
	$H = 12 - \frac{3}{100}(x - 20)^2$	A1	1.1b
	(3)		
(a) Way 3	$H = ax^2 + bx + c$ (or deduces $H = ax^2 + bx$) Both $x = 0, H = 0 \Rightarrow 0 = 0 + 0 + c \Rightarrow c = 0$ and either $x = 40, H = 0 \Rightarrow 0 = 1600a + 40b$ or $x = 20, H = 12 \Rightarrow 12 = 400a + 20b$ or $\frac{-b}{2a} = 20 \Rightarrow b = -40a$	M1	3.3
	$b = -40a \Rightarrow 12 = 400a + 20(-40a) \Rightarrow a = -0.03$ so $b = -40(-0.03) = 1.2$	dM1	3.1b
	$H = -0.03x^2 + 1.2x$	A1	1.1b
	(3)		

(b)	$\{H = 3 \Rightarrow\} 3 = \frac{3}{100}x(40 - x) \Rightarrow x^2 - 40x + 100 = 0$ or $\{H = 3 \Rightarrow\} 3 = 12 - \frac{3}{100}(x - 20)^2 \Rightarrow (x - 20)^2 = 300$	M1	3.4
	e.g. $x = \frac{40 \pm \sqrt{1600 - 4(1)(100)}}{2(1)}$ or $x = 20 \pm \sqrt{300}$	dM1	1.1b
	{chooses $20 + \sqrt{300} \Rightarrow$ } greatest distance = awrt 37.3 m	A1	3.2a
	(3)		
(c)	Gives a limitation of the model. Accept e.g. - the ground is horizontal - the ball needs to be kicked from the ground - the ball is modelled as a particle - the horizontal bar needs to be modelled as a line - there is no wind or air resistance on the ball - there is no spin on the ball - no obstacles in the trajectory (or path) of the ball - the trajectory of the ball is a perfect parabola	B1	3.5b
		(1)	

(7 marks)

Notes for Question	
(a)	
M1:	Translates the situation given into a suitable equation for the model. E.g. Way 1: {Uses (0, 0) and (40, 0) to write} $H = Ax(40 - x)$ o.e. {or $H = Ax(x - 40)$} Way 2: {Uses (20, 12) to write} $H = 12 - \lambda(x - 20)^2$ or $H = 12 + \lambda(x - 20)^2$ Way 3: Writes $H = ax^2 + bx + c$, and uses (0, 0) to deduce $c = 0$ and an attempt at using either (40, 0) or (20, 12) Special Case: Allow SC M1dM0A0 for not deducing $c = 0$ but attempting to apply both (40, 0) and (20, 12)
dM1:	Applies a complete strategy with appropriate constraints to find all constants in their model. Way 1: Uses (20, 12) on their model and finds $A = \dots$ Way 2: Uses either (40, 0) or (0, 0) on their model to find $\lambda = \dots$ Way 3: Uses (40, 0) and (20, 12) on their model to find $a = \dots$ and $b = \dots$
A1:	Finds a correct equation linking H to x E.g. $H = \frac{3}{100}x(40 - x)$, $H = 12 - \frac{3}{100}(x - 20)^2$ or $H = -0.03x^2 + 1.2x$
Note:	Condone writing y in place of H for the M1 and dM1 marks.
Note:	Give final A0 for $y = -0.03x^2 + 1.2x$
Note:	Give special case M1dM0A0 for writing down any of $H = 12 - (x - 20)^2$ or $H = x(40 - x)$ or $H = x(x - 40)$
Note:	Give M1 dM1 for finding $-0.03x^2 + 1.2x$ or $a = -0.03$, $b = 1.2$, $c = 0$ in an implied $ax^2 + bx + c$ (with no indication of $H = \dots$)
(b)	
M1:	Substitutes $H = 3$ into their quadratic equation and proceeds to obtain a 3TQ or a quadratic in the form $(x \pm \alpha)^2 = \beta$; $\alpha, \beta \neq 0$
Note:	E.g. $1.2x - 0.03x^2 = 3$ or $40x - x^2 = 100$ are acceptable for the 1 st M mark
Note:	Give M0 dM0 A0 for $(\text{their } A)x^2 = 3 \Rightarrow x = \dots$ or their $(\text{their } A)x^2 + (\text{their } k) = 3 \Rightarrow x = \dots$
dM1:	Correct method of solving their quadratic equation to give <i>at least one solution</i>
A1:	Interprets their solution in the original context by selecting the larger correct value <i>and states correct units for their value</i>. E.g. Accept awrt 37.3 m or $(20 + \sqrt{300})$ m or $(20 + 10\sqrt{3})$ m
Note:	Condone the use of inequalities for the method marks in part (b)
(c):	
B1:	See scheme
Note:	Give no credit for the following reasons H (or the height of ball) is negative when $x > 40$ - Bounce of the ball should be considered after hitting the ground - Model will not be true for a different rugby ball - Ball may not be kicked in the same way each time

Q5.

Question	Scheme	Marks	AOs
(a)	Attempts $(x-2)^2 + (y+5)^2 = \dots$	M1	1.1b
	Centre (2, -5)	A1	1.1b
		(2)	
(b)	Sets $k + 2^2 + 5^2 > 0$	M1	2.2a
	$\Rightarrow k > -29$	A1ft	1.1b
		(2)	
(4 marks)			
Notes:			
(a)			
M1: Attempts to complete the square so allow $(x-2)^2 + (y+5)^2 = \dots$			
A1: States the centre is at (2, -5). Also allow written separately $x=2, y=-5$ (2, -5) implies both marks			
(b)			
M1: Deduces that the right hand side of their $(x \pm \dots)^2 + (y \pm \dots)^2 = \dots$ is > 0 or ≥ 0			
A1ft: $k > -29$ Also allow $k \geq -29$ Follow through on their rhs of $(x \pm \dots)^2 + (y \pm \dots)^2 = \dots$			

Q6.

Question	Scheme	Marks	AOs
(a)	$u_2 = k - 12, u_3 = k - \frac{24}{k-12}$	M1	1.1b
	$u_1 + 2u_2 + u_3 = 0 \Rightarrow 2 + 2(k-12) + k - \frac{24}{k-12} = 0$	dM1	1.1b
	$\Rightarrow 3k - 22 - \frac{24}{k-12} = 0 \Rightarrow (3k-22)(k-12) - 24 = 0$ $\Rightarrow 3k^2 - 36k - 22k + 264 - 24 = 0$ $\Rightarrow 3k^2 - 58k + 240 = 0^*$	A1*	2.1
		(3)	
(b)	$k = 6, \left(\frac{40}{3}\right)$	M1	1.1b
	$k = 6$ as k must be an integer	A1	2.3
		(2)	
(c)	$(u_3 =) 10$	B1	2.2a
		(1)	
(6 marks)			
Notes			

(a)

M1: Attempts to apply the sequence formula once for either u_2 or u_3 .

Usually for $u_2 = k - \frac{24}{2}$ o.e. but could be awarded for $u_3 = k - \frac{24}{\text{their "}u_2\text{"}}$

dM1: Award for

- attempting to apply the sequence formula to find both u_2 and u_3
- using $2 + 2"u_2" + "u_3" = 0 \Rightarrow$ an equation in k . The u_3 may have been incorrectly adapted

A1*: Fully correct work leading to the printed answer.

There must be

- (at least) one correct intermediate line between $2 + 2(k - 12) + k - \frac{24}{k - 12} = 0$ (o.e.) and the given answer that shows how the fractions are "removed". E.g. $(3k - 22)(k - 12) - 24 = 0$
- no errors in the algebra. The $= 0$ may just appear at the answer line.

(b)

M1: Attempts to solve the quadratic which is implied by sight of $k = 6$.

This may be awarded for any of

- $3k^2 - 58k + 240 = (ak \pm c)(bk \pm d) = 0$ where $ab = 3, cd = 240$ followed by $k =$
- an attempt at the correct quadratic formula (or completing the square)
- a calculator solution giving at least $k = 6$

A1: Chooses $k = 6$ and gives a minimal reason

Examples of a minimal reason are

- 6 because it is an integer
- 6 because it is a whole number
- 6 because $\frac{40}{3}$ or 13.3 is not an integer

(c)

B1: Deduces the correct value of u_3 .

Q7.

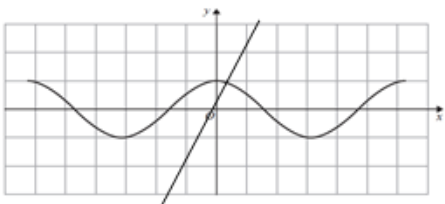
Question	Scheme	Marks	AOs
(a)	C is $(x - r)^2 + (y - r)^2 = r^2$ or $x^2 + y^2 - 2rx - 2ry + r^2 = 0$	B1	2.2a
	$y = 12 - 2x, x^2 + y^2 - 2rx - 2ry + r^2 = 0$ $\Rightarrow x^2 + (12 - 2x)^2 - 2rx - 2r(12 - 2x) + r^2 = 0$ or	M1	1.1b

	$y = 12 - 2x, (x - r)^2 + (y - r)^2 = r^2$ $\Rightarrow (x - r)^2 + (12 - 2x - r)^2 = r^2$		
	$x^2 + 144 - 48x + 4x^2 - 2rx - 24r + 4rx + r^2 = 0$ $\Rightarrow 5x^2 + (2r - 48)x + (r^2 - 24r + 144) = 0$ *	A1*	2.1
		(3)	
(b)	$b^2 - 4ac = 0 \Rightarrow (2r - 48)^2 - 4 \times 5 \times (r^2 - 24r + 144) = 0$	M1	3.1a
	$r^2 - 18r + 36 = 0$ or any multiple of this equation	A1	1.1b
	$\Rightarrow (r - 9)^2 - 81 + 36 = 0 \Rightarrow r = \dots$	dM1	1.1b
	$r = 9 \pm 3\sqrt{5}$	A1	1.1b
		(4)	
(7 marks)			

Notes:

- (a)
- B1:** Deduces the correct equation of the circle
- M1:** Attempts to form an equation with terms of the form x^2 , x , r^2 , and rx only using $y = 12 \pm 2x$ and their circle equation which must be of an appropriate form. I.e. includes or implies an x^2 , y^2 , r^2 such as $x^2 + y^2 = r^2$
- If their circle equation starts off as e.g. $(x \pm a)^2 + (y \pm b)^2 = r^2$ then the B mark and the M mark can be awarded when the "a" and "b" are replaced by r or $-r$ as appropriate for their circle equation.
- A1*:** Uses correct and accurate algebra leading to the given solution.
- (b)
- M1:** Attempts to use $b^2 - 4ac \dots 0$ o.e. with $a = 5, b = 2r - 48, c = r^2 - 24r + 144$ and where ... is "=" or any inequality
- Allow minor slips when copying the a, b and c provided it does not make the work easier and allow their a, b and c if they are similar expressions.
- FYI $(2r - 48)^2 - 4 \times 5 \times (r^2 - 24r + 144) = 4r^2 - 192r + 2304 - 20r^2 + 480r - 2880 = -16r^2 + 288r - 576$
- A1:** Correct quadratic equation in r (or inequality). Terms need not be all one side but must be collected.
- E.g. allow $r^2 - 18r = -36$ and allow any multiple of this equation (or inequality).
- dM1:** Correct attempt to solve their 3TQ in r . Dependent upon previous M
- A1:** Careful and accurate work leading to both answers in the required form (must be simplified surds)

Q8.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)		M1	This mark is given for plotting the line $y = 2x + \frac{1}{2}$ on the diagram with a correct gradient and intercept
	Only one intersection means that there is one root	A1	This mark is given for a reason why there is only one real root
(b)	$1 - \frac{x^2}{2} - 2x - \frac{1}{2} = 0$	M1	This mark is given for using the small angle approximation $\cos x = 1 - \frac{x^2}{2}$ in the given equation
	$x^2 + 4x - 1 = 0$	M1	This mark is given for rearranging to find a quadratic equation to solve
	0.236 or $-2 + \sqrt{5}$	A1	This mark is given for finding the correct (positive) solution for x

Q9.

Question	Scheme	Marks	AOs
(a)	$H = ax^2 + bx + c$ and $x=0, H=3 \Rightarrow H = ax^2 + bx + 3$	M1	3.3
	$H = ax^2 + bx + 3$ and $x=120, H=27 \Rightarrow 27 = 14400a + 120b + 3$	M1	3.1b
	or $\frac{dH}{dx} = 2ax + b = 0$ when $x = 90 \Rightarrow 180a + b = 0$	A1	1.1b
	$H = ax^2 + bx + 3$ and $x=120, H=27 \Rightarrow 27 = 14400a + 120b + 3$ and $\frac{dH}{dx} = 2ax + b = 0$ when $x = 90 \Rightarrow 180a + b = 0$ $\Rightarrow a = \dots, b = \dots$	dM1	3.1b
	$H = -\frac{1}{300}x^2 + \frac{3}{5}x + 3$ o.e.	A1	1.1b
		(5)	
(b)(i)	$x = 90 \Rightarrow H \left(= -\frac{1}{300}(90)^2 + \frac{3}{5}(90) + 3 \right) = 30 \text{ m}$	B1	3.4
(b)(ii)	$H = 0 \Rightarrow -\frac{1}{300}x^2 + \frac{3}{5}x + 3 = 0 \Rightarrow x = \dots$	M1	3.4
	$x = (-4.868\dots) 184.868\dots$ $\Rightarrow x = 185 \text{ (m)}$	A1	3.2a
		(3)	
(c)	Examples must focus on why the model may not be appropriate or give values/situations where the model would break down: E.g. - The ground is unlikely to be horizontal - The ball is not a particle so has dimensions/size - The ball is unlikely to travel in a vertical plane (as it will spin) H is not likely to be a quadratic function in x	B1	3.5b
		(1)	
(9 marks)			
Notes			

(a)

M1: Translates the problem into a suitable model and uses $H = 3$ when $x = 0$ to establish $c = 3$

Condone with $a = \pm 1$ so $H = x^2 + bx + 3$ will score M1 but little else

M1: For a correct attempt at using one of the two other pieces of information within a quadratic model

Either uses $H = 27$ when $x = 120$ (with $c = 3$) to produce a linear equation connecting a and b for

the model Or differentiates and uses $\frac{dH}{dx} = 0$ when $x = 90$. Alternatives exist here, using the

symmetrical nature of the curve, so they could use $x = -\frac{b}{2a}$ at vertex or use point (60, 27) or (180, 3).

- A1: At least one correct equation connecting a and b . Remember " a " could have been set as negative so an equation such as $27 = -14400a + 120b + 3$ would be correct in these circumstances.
- dM1: Fully correct strategy that uses $H = ax^2 + bx + 3$ with the two other pieces of information in order to establish the values of both a and b for the model
- A1: Correct equation, not just the correct values of a , b and c . Award if seen in part (b)
- (b)(i)
- B1: Correct height including the units. CAO
- (b)(ii)
- M1: Uses $H = 0$ and attempts to solve for x . Usual rules for quadratics.
- A1: Discards the negative solution (may not be seen) and identifies awrt 185 m. Condone lack of units
- (c)
- B1: Candidate should either refer to an issue with one of the four aspects of how the situation has been modelled or give a situation where the model breaks down
- the ball has been modelled as a particle
 - there may be trees (or other hazards) in the way that would affect the motion
- Condone answers (where the link to the model is not completely made) such as
- the ball will spin
 - ground is not flat
- Do not accept answers which refer to the situation after it hits the ground (this isn't what was modelled)
- the ball will bounce after hitting the ground
 - it gives a negative height for some values for x
- Do not accept answers that do not refer to the model in question, or else give single word vague answers
- the height of tee may have been measured incorrectly
 - "friction", "spin", "force" etc
 - it does not take into account the weight of the ball
 - it depends on how good the golfer is
 - the shape of the ball will affect the motion
 - you cannot hit a ball the same distance each time you hit it

The method using an alternative form of the equation can be scored in a very similar way.

The first M is for the completed square form of the quadratic showing a maximum at $x = 90$

So award M1 for $H = \pm a(x-90)^2 + c$ or $H = \pm a(90-x)^2 + c$. Condone for this mark an equation with

$a = 1 \Rightarrow H = (x-90)^2 + c$ or $c = 3 \Rightarrow H = a(x-90)^2 + 3$ but will score little else

Alt (a)	$H = a(x+b)^2 + c$ and $x=90$ at $H_{\max} \Rightarrow H = a(x-90)^2 + c$	M1	3.3
	$H = 3$ when $x = 0 \Rightarrow 3 = 8100a + c$	M1	3.1b
	or	A1	1.1b
	$H = 27$ when $x = 120 \Rightarrow 27 = 900a + c$		
	$H = 3$ when $x = 0 \Rightarrow 3 = 8100a + c$ and $H = 27$ when $x = 120 \Rightarrow 27 = 900a + c$ $\Rightarrow a = \dots, c = \dots$	dM1	3.1b
	$H = -\frac{1}{300}(x-90)^2 + 30$ o.e	A1	1.1b
		(5)	
(b)	$x = 90 \Rightarrow H = 0^2 + 30 = 30$ m	B1	3.4
		(1)	
	$H = 0 \Rightarrow 0 = -\frac{1}{300}(x-90)^2 + 30 \Rightarrow x = \dots$	M1	3.4
	$\Rightarrow x = 185$ (m)	A1	3.2a
		(2)	

Note that $H = -\frac{1}{300}(x-90)^2 + 30$ is equivalent to $H = -\frac{1}{300}(90-x)^2 + 30$

Other versions using symmetry are also correct so please look carefully at all responses

E.g. Using a starting equation of $H = a(x-60)(x-120) + b$ leads to $H = -\frac{1}{300}(x-60)(x-120) + 27$

Q10.

Question	Scheme	Marks	AOs
(a)	$(1+8x)^{\frac{1}{2}} = 1 + \frac{1}{2} \times 8x + \frac{\frac{1}{2} \times -\frac{1}{2}}{2!} \times (8x)^2 + \frac{\frac{1}{2} \times -\frac{1}{2} \times -\frac{3}{2}}{3!} \times (8x)^3$	M1 A1	1.1b 1.1b
	$= 1 + 4x - 8x^2 + 32x^3 + \dots$	A1	1.1b
		(3)	
(b)	Substitutes $x = \frac{1}{32}$ into $(1+8x)^{\frac{1}{2}}$ to give $\frac{\sqrt{5}}{2}$	M1	1.1b
	Explains that $x = \frac{1}{32}$ is substituted into $1 + 4x - 8x^2 + 32x^3$ and you multiply the result by 2	A1ft	2.4
		(2)	
(5 marks)			
Notes:			

(a)

M1: Attempts the binomial expansion with $n = \frac{1}{2}$ and obtains the correct structure for term 3 or term 4.

Award for the correct coefficient with the correct power of x . Do not accept nC_r notation for coefficients.

For example look for term 3 in the form $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times (*x)^2$ or $\frac{1}{2} \times \frac{1}{2} \times \frac{3}{2} \times (*x)^3$

A1: Correct (unsimplified) expression. May be implied by correct simplified expression

A1: $1 + 4x - 8x^2 + 32x^3$

Award if there are extra terms (even if incorrect).

Award if the terms are listed $1, 4x, -8x^2, 32x^3$

(b)

M1: Score for substituting $x = \frac{1}{32}$ into $(1+8x)^{\frac{1}{2}}$ to obtain $\frac{\sqrt{5}}{2}$ or equivalent such as $\sqrt{\frac{5}{4}}$

Alternatively award for substituting $x = \frac{1}{32}$ into **both** sides and making a connection between the two sides by use of an $=$ or $\approx$.

E.g. $\left(1 + \frac{8}{32}\right)^{\frac{1}{2}} = 1 + 4 \times \frac{1}{32} - 8 \times \left(\frac{1}{32}\right)^2 + 32 \times \left(\frac{1}{32}\right)^3$ following through on their expansion

Also implied by $\frac{\sqrt{5}}{2} = \frac{1145}{1024}$ for a correct expansion

It is not enough to state substitute $x = \frac{1}{32}$ into "the expansion" or just the rhs " $1 + 4x - 8x^2 + 32x^3$ "

A1ft: Requires a full (and correct) **explanation** as to how the expansion can be used to estimate $\sqrt{5}$

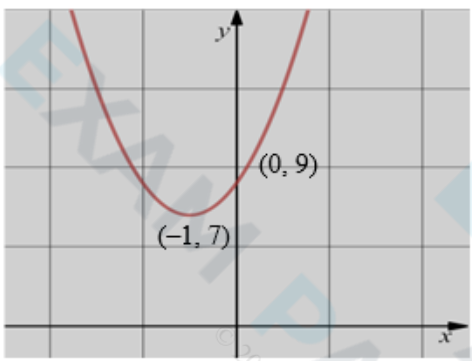
E.g. Calculates $1 + 4 \times \frac{1}{32} - 8 \times \left(\frac{1}{32}\right)^2 + 32 \times \left(\frac{1}{32}\right)^3$ and multiplies by 2.

This can be scored from an incorrect binomial expansion or a binomial expansion with more terms.

The explanation could be mathematical. So $\frac{\sqrt{5}}{2} = \frac{1145}{1024} \rightarrow \sqrt{5} = \frac{1145}{512}$ is acceptable.

SC: For 1 mark, M1, A0 score for a statement such as "substitute $x = \frac{1}{32}$ into both sides of part (a) and make $\sqrt{5}$ the subject"

Q11.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$2x^2 + 4x + 9 = 2(x + b)^2 + c$	B1	This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$
	$2x^2 + 4x + 9 = 2(x + 1)^2 + c$	M1	This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$ and $b = 1$
	$2x^2 + 4x + 9 = 2(x + 1)^2 + 7$	A1	This mark is given for writing $f(x)$ in the form $a(x + b)^2 + c$ with $a = 2$, $b = 1$ and $c = 7$
(b)		B1	This mark is given for a U shaped curve in any position
		B1	This mark is given for a y-intercept shown at (0, 9)
		B1	This mark is given for a minimum shown at (-1, 7)
(c)(i)	$g(x) = 2(x - 2)^2 + 4(x - 2) + 5$	M1	This mark is given for writing $g(x)$ in the form $a(x + b)^2 + c$ and comparing to $f(x)$
	Translation of $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$	A1	This mark is given for deducing the translation of $y = f(x)$ to $y = g(x)$
(c)(ii)	$h(x) = \frac{21}{2(x+1)^2 + 7}$ Maximum value = $\frac{21}{7}$ (when $x = -1$)	M1	This mark is given for writing $h(x)$ in the form $\frac{21}{a(x+b)^2 + c}$ and finding its maximum value
	$0 < h(x) \leq 3$	A1	This mark is given for finding the correct range of the function $h(x)$

Q12.

Question	Scheme	Marks	AOs
(a)	$f(x) = (x - 2)^2 \pm \dots$	M1	1.2
	$f(x) = (x - 2)^2 + 1$	A1	1.1b
		(2)	
(b)(i)	$P = (0, 5)$	B1	1.1b
(b)(ii)	$Q = (2, 1)$	B1ft	1.1b
		(2)	
(4 marks)			
Notes			

(a)

M1: Achieves $(x-2)^2 \pm \dots$ or states $a = -2$

A1: Correct expression $(x-2)^2 + 1$ ISW after sight of this

Condone $a = -2$ and $b = 1$. Condone $(x-2)^2 + 1 = 0$

(b)

(i) B1: Correct coordinates for P . Allow to be expressed $x = 0, y = 5$

(ii) B1ft: Correct coordinates for Q . Allow to be expressed $x = 2, y = 1$ (Score for the correct answer or follow through their part (a) so allow $(-a, b)$ where a and b are numeric)

Score in any order if they state $P = (0, 5)$ and $Q = (2, 1)$

.....
Allow part (b) to be awarded from a sketch. So award

First B1 from a sketch crossing the y -axis at 5

Second B1 from a sketch with minimum at $(2, 1)$
.....

Q13.

Question	Scheme	Marks	AOs
(a)	Uses $\cos^2 x = 1 - \sin^2 x \Rightarrow 3 \sin^2 x + \sin x + 8 = 9(1 - \sin^2 x)$	M1	3.1a
	$\Rightarrow 12 \sin^2 x + \sin x - 1 = 0$	A1	1.1b
	$\Rightarrow (4 \sin x - 1)(3 \sin x + 1) = 0$	M1	1.1b
	$\Rightarrow \sin x = \frac{1}{4}, -\frac{1}{3}$	A1	1.1b
	Uses arcsin to obtain two correct values	M1	1.1b
	All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$	A1	1.1b
		(6)	
(b)	Attempts $2\theta - 30^\circ = -19.47^\circ$	M1	3.1a
	$\Rightarrow \theta = 5.26^\circ$	A1ft	1.1b
		(2)	
(8 marks)			
Notes:			
(a) M1: Substitutes $\cos^2 x = 1 - \sin^2 x$ into $3 \sin^2 x + \sin x + 8 = 9 \cos^2 x$ to create a quadratic equation in just $\sin x$ A1: $12 \sin^2 x + \sin x - 1 = 0$ or exact equivalent M1: Attempts to solve their quadratic equation in $\sin x$ by a suitable method. These could include factorisation, formula or completing the square. A1: $\sin x = \frac{1}{4}, -\frac{1}{3}$ M1: Obtains two correct values for their $\sin x = k$ A1: All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$			
(b) M1: For setting $2\theta - 30^\circ = \text{their } -19.47^\circ$ A1ft: $\theta = 5.26^\circ$ but allow a follow through on their -19.47°			

Q14.

Question	Scheme	Marks	AOs
(a)	Sets $H = 0 \Rightarrow 1.8 + 0.4d - 0.002d^2 = 0$	M1	3.4
	Solves using an appropriate method, for example $d = \frac{-0.4 \pm \sqrt{(0.4)^2 - 4(-0.002)(1.8)}}{2 \times -0.002}$	dM1	1.1b
	Distance = awrt 204(m) only	A1	2.2a
		(3)	
(b)	States the initial height of the arrow above the ground.	B1	3.4
		(1)	
(c)	$1.8 + 0.4d - 0.002d^2 = -0.002(d^2 - 200d) + 1.8$	M1	1.1b
	$= -0.002((d - 100)^2 - 10000) + 1.8$	M1	1.1b
	$= 21.8 - 0.002(d - 100)^2$	A1	1.1b
		(3)	
(d)	(i) 22.1 metres	B1ft	3.4
	(ii) 100 metres	B1ft	3.4
		(2)	
(9 marks)			

Notes:

(a)

M1: Sets $H = 0 \Rightarrow 1.8 + 0.4d - 0.002d^2 = 0$

M1: Solves using formula, which if stated must be correct, by completing square (look for $(d - 100)^2 = 10900 \Rightarrow d = \dots$) or even allow answers coming from a graphical calculator

A1: Awrt 204 m only

(b)

B1: States it is the initial height of the arrow above the ground. Do not allow "it is the height of the archer"

(c)

M1: Score for taking out a common factor of -0.002 from at least the d^2 and d terms

M1: For completing the square for their $(d^2 - 200d)$ term

A1: $= 21.8 - 0.002(d - 100)^2$ or exact equivalent

(d)

B1ft: For their '21.8+0.3' = 22.1m

B1ft: For their 100m

Q15.

Question	Scheme	Marks	AOs
	$\int (3x^{0.5} + A) dx = 2x^{1.5} + Ax(+c)$	M1 A1	3.1a 1.1b
	Uses limits and sets $= 2A^2 \Rightarrow (2 \times 8 + 4A) - (2 \times 1 + A) = 2A^2$	M1	1.1b
	Sets up quadratic and attempts to solve Sets up quadratic and attempts $b^2 - 4ac$	M1	1.1b
	$\Rightarrow A = -2, \frac{7}{2}$ and states that there are two roots States $b^2 - 4ac = 121 > 0$ and hence there are two roots	A1	2.4
(5 marks)			
Notes:			
M1: Integrates the given function and achieves an answer of the form $kx^{1.5} + Ax(+c)$ where k is a non- zero constant A1: Correct answer but may not be simplified M1: Substitutes in limits and subtracts. This can only be scored if $\int A dx = Ax$ and not $\frac{A^2}{2}$ M1: Sets up quadratic equation in A and either attempts to solve or attempts $b^2 - 4ac$ A1: Either $A = -2, \frac{7}{2}$ and states that there are two roots Or states $b^2 - 4ac = 121 > 0$ and hence there are two roots			